

Prayas JEE 2025

Maths

DPP: 4

Sequence and Series

- Q1** If a, b, c are distinct and are in G.P with common ratio r and $a, 2b, 3c$ are in AP then $r =$
 (A) 1
 (B) $\frac{1}{3}$
 (C) $-\frac{1}{3}$
 (D) -1
- Q2** If $a_1 a_2, \dots, a_{10}$ are in G.P and $\frac{a_3}{a_1} = 25$ then $\frac{a_9}{a_5} =$
 (A) $2(5^2)$
 (B) $4(5^2)$
 (C) 5^4
 (D) 5^3
- Q3** The number which should be added to the numbers 2, 14, 62 so that resulting numbers are in GP is
 (A) 1
 (B) 2
 (C) 3
 (D) 4
- Q4** Fifth term of a GP is 2 then the product of its first 9 terms is
 (A) 256
 (B) 512
 (C) 1024
 (D) None
- Q5** If a, b, c are three unequal numbers such that a, b, c are in AP and $b - a, c - b, a$ are in GP then $a : b : c =$
 (A) 1 : 2 : 3
 (B) 1 : 3 : 4
 (C) 2 : 3 : 4
 (D) 1 : 2 : 4
- Q6** If $p^{\text{th}}, q^{\text{th}},$ and r^{th} term of a GP are again in GP then $p, q, r,$ are in
 (A) AP
 (B) GP
 (C) HP
 (D) None
- Q7** If the second and fifth terms of a GP are 24 and 3 respectively, then the sum of first six terms is
 (A) 181
 (B) $\frac{181}{2}$
 (C) 189
 (D) $\frac{189}{2}$
- Q8** The third term of a G.P. is 3. The product of its first five terms is
 (A) 81
 (B) 243
 (C) 729
 (D) 343
- Q9** If the sixth term of a GP be 2, then the product of first eleven terms is
 (A) 1024
 (B) 2047
 (C) 2048
 (D) 1023
- Q10** The fifth terms of a GP. is 32 and common ratio is 2, then the sum of first 14 terms of the G.P. is
 (A) 16388
 (B) 32667
 (C) 32766
 (D) 64342
- Q11** If the sum of first three numbers in GP is 21 and their product is 216, then the numbers are
 (A) 3, 6, 12
 (B) 5, 7, 9
 (C) 6, 2, 213
 (D) 6, 12, 24



Q12 The n^{th} terms of a GP is 128 and the sum of its n terms is 255. If its common ratio is 2 then find the first term.

- (A) 1 (B) 2
(C) 3 (D) 4



Answer Key

Q1 (B)
Q2 (C)
Q3 (B)
Q4 (B)
Q5 (A)
Q6 (A)

Q7 (D)
Q8 (B)
Q9 (C)
Q10 (C)
Q11 (A)
Q12 (A)



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